

A Gamified Approach for Screening and Intervention of Dyslexia, Dysgraphia and Dyscalculia

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Abstract—This paper aims to diagnose children with specific learning disabilities and provide treatments via a mobile game. Learning disabilities are neurological disorders that affect the brain. Children with learning disabilities have trouble with learning compared to their fellow peers and quite often fall back academically since a majority of them go undiagnosed. The specific learning disabilities for which this paper provides screening are dyslexia a reading disability, dyscalculia a mathematical disability, letter dysgraphia and numeric dysgraphia are both writing disabilities. Deep learning and machine learning techniques are used in the screening process of these specific learning disabilities. Trained convolutional neural networks are used to detect the spoken letter/word, detect the written letter/word and detect the written number on the mobile application. Outputs from the convolutional neural network are fed into the models used for screening learning disabilities. The machine learning algorithms used in building the models include k-nearest neighbors, random forest and support vector machine. Screening results from the models built in this research provided an accuracy of 89%, 90%, 92%, 92% for dyslexia, letter dysgraphia, dyscalculia and numeric dysgraphia respectively. This is the first game based screening and intervention tool for dyslexia, letter dysgraphia, dyscalculia and numeric dysgraphia.

Keywords—*Learning disabilities, Dyslexia, Dysgraphia, Dyscalculia, Neural network, Machine learning, Mobile application*

I. INTRODUCTION

A learning disability is a neurological disorder or simply described as a difference in the structure of the brain[1]. Children with learning disabilities may have difficulty in reading, which is called dyslexia, some clinical symptoms of dyslexia being mispronunciation of words, slow reading and reading ability below the expected age. It is estimated that at least 10% of the world's population is dyslexic [2]. Difficulty in writing is a specific learning disability called dysgraphia which is characterised by symptoms such as poor handwriting, uneven spacing between words, varying letter sizes and the inability to stay within line margins [3]. Approximately 10% - 30% of schooling children are estimated to be affected by dysgraphia [4]. Dyscalculia is difficulty in learning arithmetic with symptoms being difficulty in counting, number comparison and arithmetic addition [5]. It is estimated that around 3% - 6% of schooling children are affected by this learning disability.

These specific learning disabilities cannot be completely cured by medications. Treatment of these learning disabilities consists of providing extra training to overcome the symptoms. In developing countries such as Sri Lanka, these learning

disabilities that exist among school-going children often go undiagnosed due to a lack of awareness. Also, currently existing screening methods consist of paper-based tests which are difficult and time-consuming to administer. The aim of this research is to develop an application having models that are capable of detecting dyslexia, dysgraphia and dyscalculia among the age groups of 6-7 and provide the necessary therapy to alleviate symptoms. Section II discusses the literature survey. Section III explains the methodology used in making the system. Section IV gives the system results based on accuracy. Section V discusses the conclusion.

II. LITERATURE REVIEW

Dyslexia has been defined as “a specific language-based disorder of constitutional origin characterised by difficulties in single word decoding, usually reflecting insufficient phonological processing” by Dr. Reid Lyon [6]. In studies [6-7] they have identified five most common symptoms of dyslexic people as, word confusion, word processing, spelling, phonology and short-term memory. Moreover, they have estimated that 10% - 17% of school children are dyslexic around the world and found that computer-based learning models are very attractive and user-friendly over traditional methods [7]. Researches [7-9] have emphasized a creative learning technique to help dyslexics by providing guidelines for web texts, frequent words and how graphical schema affects dyslexics.

In considering the existing applications in Sri Lanka, [10] was the first application to perform screening of dyslexia, dysgraphia and dyscalculia utilizing tablet computer game series having limitations to use the Kinect device. The purpose of the study [11] is to conduct online screening of dyslexia followed by a questionnaire, but it is designed only for young adults. A study in [12] done for screening dyslexia using computational analysis with 96% accuracy, but it lacks language portability.

Research in [13] has designed a user-friendly mobile application to assist children to recognize, memorize and repeat letters but this study does not support dyslexia screening. A gamified environment for improving reading skills through playing action games was provided in [14] for preschoolers but the drawback was that it was designed for preschoolers. A study in [15] was designed to assist people with dyslexia, inspire them to overcome difficulties in existing e-reading apps with a prototype consisting of new features. Less user-friendliness and age group limitation are the major issues in this study. In the study [16] they have proven that the use of gamification can be useful to motivate the dyslexic community. The study in [17] is

a text to speech online learning tool for reading with text magnifying features but it lacks dictionary support and text customization features.

In [18], a soft computing approach was proposed for diagnosing and classification of learning disability. The model was designed by implementing a soft computing technique called learning vector quantization which classifies a child as learning disabled or not learning disabled. Once diagnosed with a learning disability, a rule-based approach is further used to classify them into types of learning disability that is dyslexia, dysgraphia and dyscalculia. Studies [19-21] have emphasized the benefits of computerized systems for the identification of dysgraphia features on handwriting. Handwriting screening questionnaire for children aged from 7 -14 was conducted in [22] with limitations of having a convenience sample and limitations in language portability. Out of the few screening applications for dysgraphia [23] was based on forward-chaining with a good accuracy level, furthermore [24] was a study done for diagnosis of dysgraphia using handwriting classification methods to develop a statistical model for discriminating between dysgraphic and proficient handwriting with an accuracy of 90% but the limitation was less user-friendliness as a result parent guidance was a must. In the study [25] two short and practical questionnaires were used to evaluate dysgraphia with an accuracy of 96% but it doesn't support language portability and with a 10% handwriting proficiency screening questionnaire estimation error. Dysgraphicoach is a mobile application to support dysgraphic children using Malaysian language with different colours and fonts to improve writing skills but it is limited to the Android operating system [26].

A series of games were used in [27] for dysgraphia intervention in an organized manner to trace and write English letters. Study [28] was done to identify lifts and stops in proficient and dysgraphic handwriting in Roman writing for adults and children using tablets in a less user-friendly manner. In the study [29] it provides augmented reality-based dysgraphia assistive writing environment to address the spelling mistakes.

But it will yield good results for phonological dysgraphic students only. One research carried out, proposed an augmented reality-based spelling assistance app for students with dysgraphia [30] which uses voice to text technologies. The drawback is that this app addresses only the spelling issue of dysgraphic students.

While dyscalculia has many symptoms, some of the symptoms that are shown earliest in a child with dyscalculia is the difficulty of enumerating or counting between the range of one to nine [31], difficulty in discriminating which number is largest when given a series of numbers and basic arithmetic skills such as addition [32].

Reviewing other research, DyscalculiUM [33] was designed as a standalone computer-based system for undergraduates and does not support screening of primary age school children, while [34] is a screening test designed only for use on Android tablets and [35] is a computer-based screener that is not very attractive to be used by children. Another web application, Dyscalc [36] is a free online screening instrument developed for students above the age of 14. Furthermore [37] is a two-minute paper-and-pencil assessment for screening children's early mathematical performance, this assessment is purely paper-based. A research [38] based on classifying dyscalculia found that SVM was among the most accurate classifiers for dyscalculia.

For the dyscalculia intervention, [39] is a tool designed to improve the performance of students with dyscalculia, but it is a standalone system and not a mobile application while [40] is a web-based learning tool but neither of them supports local languages. Moreover, [41] is a computer application developed only for students between the age of 13 to 17 to improve their mathematical skills while our system provides intervention targeting a much earlier age of 6 to 7 years. MathBoard [42], is an iPad only application that helps kindergarten children in learning mathematics.

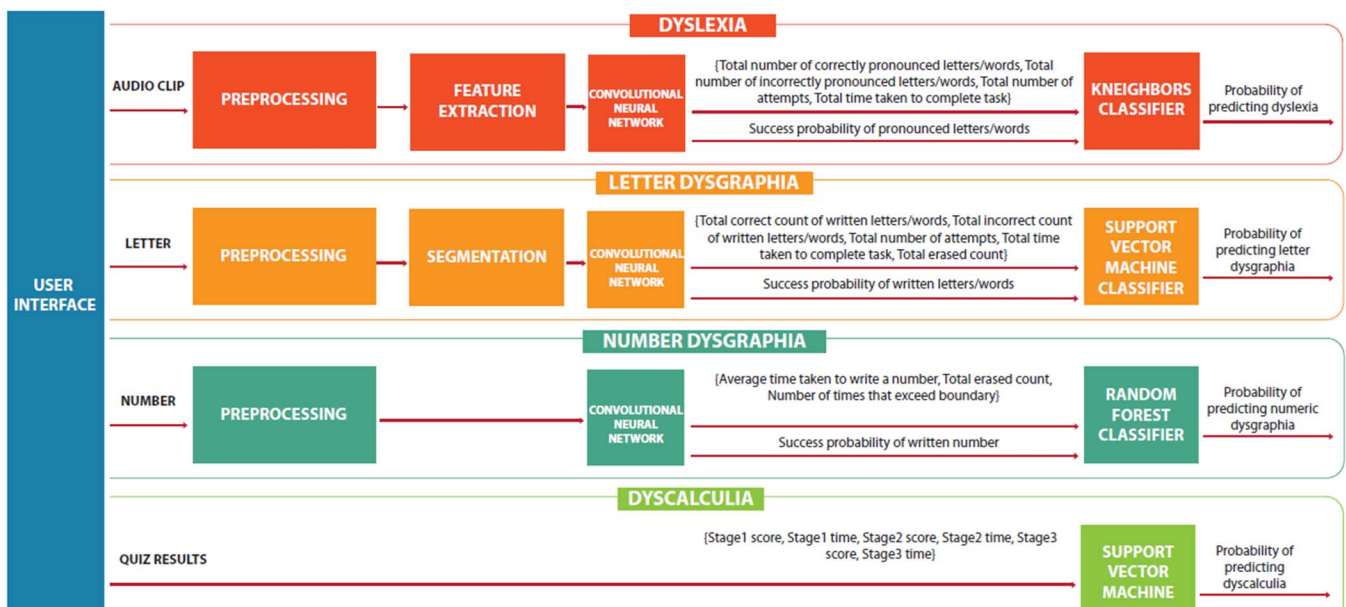


Fig. 1. System Diagram

III. METHODOLOGY

Four different strategies are adopted for the screening and intervention of dyslexia, dyscalculia and dysgraphia for letters and numbers. Fig. 1 shows the system diagram.

A. Screening and intervention of dyslexia

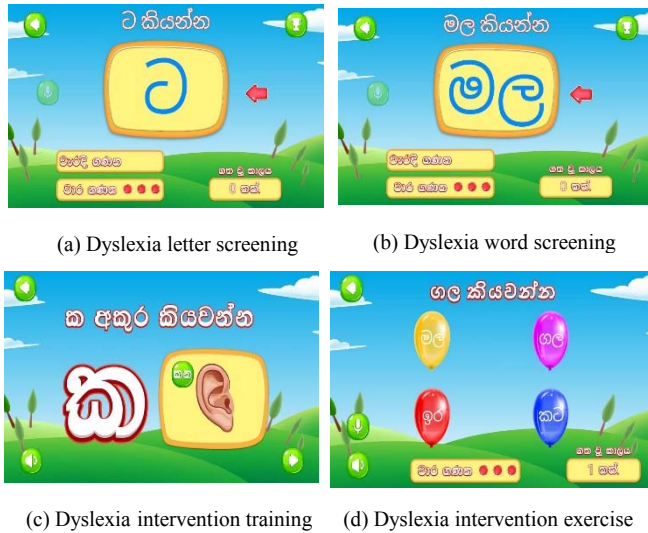


Fig. 2. Sample Dyslexia Interfaces

A series of pronunciation tests were carried out to validate dyslexic and non-dyslexic children. As shown in Fig. 2.a initially users were asked to pronounce the word displayed on the mobile screen and then gradually the level of complexity was increased to pronounce a word as shown in Fig. 2.b. The model was trained with 1000 audio clips for each word /letter with a 0.8-second duration. Preprocessing was done to eliminate unwanted frequencies as recordings were gathered in the natural environment of children. A zero padding technique was used to prepare a fixed audio clip. Mel-Frequency Cepstral Coefficient (MFCC) yielded precise results for feature extraction. Convolutional Neural Network (CNN) based acoustic model yielded better accuracy for classifying probabilities of phoneme classes than Recurrent Neural Network (RNN). Softmax yielded good accuracy in comparison with LeakyRelu, furthermore, the number of epochs was increased to 150 to gain a high accuracy level. Five parameters were fed into the model during screening namely success probability of pronounced letters/words, total number of correctly pronounced letters/words, total number of incorrectly pronounced letters/words, the user is dyslexic based on K-Nearest Neighbors algorithm (KNN).

If the child was predicted as dyslexic, a multisensory gaming environment was created for intervention. It was further divided into two categories namely easy and hard. In easy category, users were tested and trained on single letters while in hard stages users were trained and tested on two letters. Graphical card interfaces were provided for training as shown in Fig. 2.c to recall words and letters through pictures. In the easy stage, a simple exercise was provided to select a corresponding letter represented on a balloon. In the hard stage, users were asked to pronounce the word displayed as shown in Fig. 2.d. If he/she pronounced the word correctly the balloon will burst with a

congratulating screen. The competency level was measured based on total time taken and the total number of attempts taken in interventions. If intervention mechanisms were satisfactory then the users will be navigated back to screening mechanisms; if not users are retained in the intervention stage to practice more.

B. Screening and intervention of letter dysgraphia



Fig. 3. Sample Letter Dysgraphia Interfaces

In dysgraphia screening, a series of writing tests were carried out; initially, children were asked to write letters on a mobile screen with the help of a pen as shown in Fig. 3.a and later the users were asked to write words on a mobile screen.

2500 images were collected from normal children for each letter. Images were preprocessed to remove noise after conversion to binary. The images which were normalized and preprocessed were then separated into individual letters using the segmentation process. Multi-class cross entropy was the loss function and a CNN with 8 layers was used.

Support Vector Machine (SVM) algorithm was used to predict whether users are having dysgraphia or not. Training dataset for classifying letter dysgraphia comprised of 100 cases. 50 cases of children that are affected by letter dysgraphia and 50 cases of normal children without letter dysgraphia.

Six inputs fed to the SVM model were success probability of the written letters/words, total correct count of written letters/words, total incorrect count of written letters/words, total number of attempts, total time taken to complete the task and total erased count in screening mechanism. The SVM model would then classify the child as normal or affected by letter dysgraphia.

Navigation to intervention was done if the child was predicted as dysgraphic. Initially, dysgraphic children will be trained to write letters in the proceeding path with precise start and endpoints in a defined scope with the help of animations as shown in Fig. 3.b. Users were allowed to write letters independently within the given scope in later stages of intervention. Users were allowed to write two letters in a given

scope and the total number of nodes covered and the proceeding path of letters were taken into consideration in deciding the level of competency as shown in Fig. 3.c. If intervention mechanisms were satisfactory then users will be rescreened after displaying the summary of intervention results as shown in Fig. 3.d.

C. Screening and intervention of dyscalculia

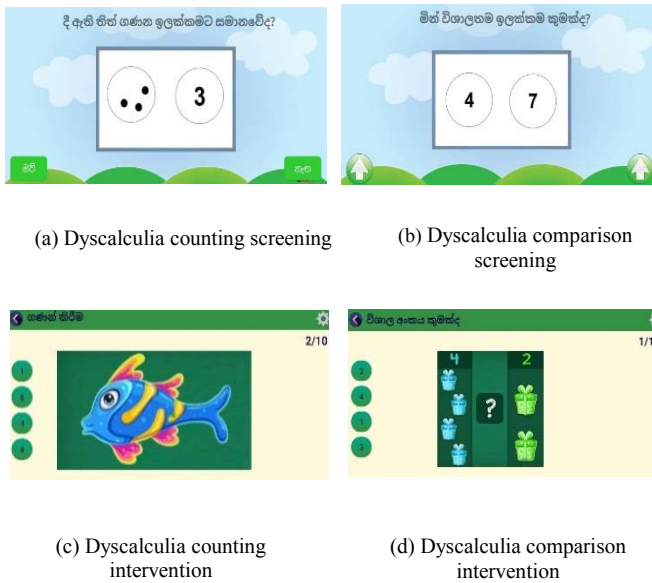


Fig. 4. Sample Dyscalculia Interfaces

The screening process for dyscalculia involves administering a mathematical test that checks the mathematical ability of a school child. Questions for the test are stored in a question bank and the questions are chosen randomly during the screening test. The skill areas which are tested in the screening of dyscalculia are counting, number comparison and arithmetic addition. The screening process was built with the advice of medical professionals so that the test is capable of screening children for dyscalculia within the age of 6 to 7.

The dyscalculia screening tests are split into the following three stages. The first stage checks the counting ability as shown in Fig. 4.a, and the second stage checks the number comparison ability as shown in Fig. 4.b. In the final stage, basic arithmetic addition is checked. A model was created with 6 inputs being counting accuracy, time spent per question in counting stage, number comparison accuracy, time spent per question in number comparison stage, arithmetic addition accuracy, time spent per question in the addition stage. An SVM classifier with a Radial Basis Function (RBF) kernel was used to build the model which classifies children as affected by dyscalculia or not.

The training dataset consists of 100 cases, out of which 50 are of normal children and the remaining 50 cases consist of children with dyscalculia.

Based on the results of this screening process, if a child is identified as having dyscalculia, the child will be directed to the intervention activities which graphically teaches math problems. Intervention provides training for counting as shown in Fig. 4.c, training in number comparison as shown in Fig. 4.d and training in arithmetic addition. The intervention game built uses a

question bank so that each new attempt at the counting, number comparison and arithmetic addition skill stages begins with new questions.

D. Screening and intervention of numeric dysgraphia

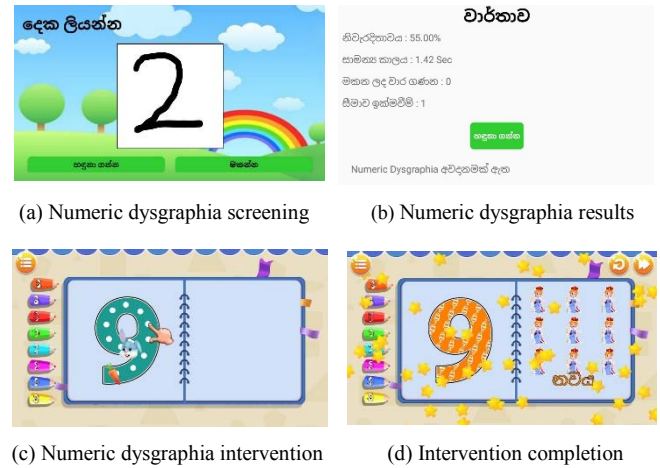


Fig. 5. Sample Numeric Dysgraphia Interfaces

The screening process of numeric dysgraphia starts from a number writing quiz as shown in Fig. 5.a. Initially, a user is asked to write numbers from 0 to 9, twice on the canvas provided. To train the model to predict numeric dysgraphia four inputs were fed, namely total erased count, number of times out of boundary, the success probability of written numbers and the average time taken to complete the tasks. In training the model, a group of 100 children, 50 pre-diagnosed with dysgraphia and 50 typical children were taken.

The evaluation of the test is carried out using a Random Forest (RF) classifier. Upon completion of the screening test, test results will be displayed as shown in Fig. 5. b and by clicking on the green button, the user is notified whether he/she is dysgraphic or not. In identifying the correctness of the number Modified National Institute of Standards and Technology (MNIST) numbers dataset is used. The CNN consists of 19 layers and 20 epochs were used to obtain a high accuracy from the CNN. The numbers model was then converted to a TensorFlow Lite file.

If a child is predicted dysgraphic according to the screening test, he/she will be navigated to the intervention as shown in Fig. 5.c. This is provided in a multisensory gaming environment where a user has to trace the given number with the help of animations. For each number, graphical objects which are equal to the number count will be displayed after the completion as shown in Fig. 5.d. The intervention is designed in a way that it develops observation and motor skills of the dysgraphic children after getting consultations from medical and educational professionals. Parents can view the screening results of their children by viewing the screening report.

IV. RESULTS

The application was tested among pre-diagnosed children with learning disabilities studying in special units and Senehasa Education Resource Research and Information Centre (SERRIC) and the results obtained reached a high accuracy.

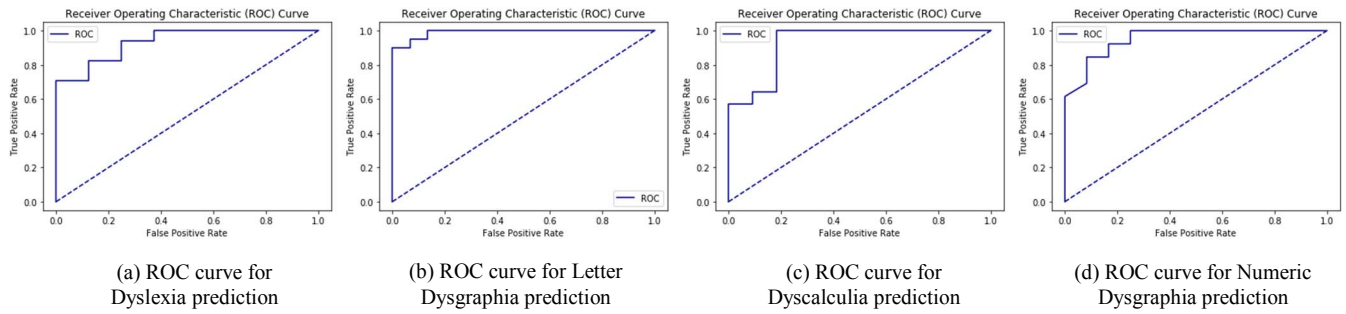


Fig. 6. ROC curves for predicting each learning disability

Testing the interventions for these learning disabilities is still in progress as children should be given at least six months period training through the application according to the recommendations of health and educational professionals.

A. Dyslexia screening

Screening was done for six Sinhala letters, for each letter 1000 audio clips with 0.8 second duration were gathered from normal school children within age six to seven. CNN yielded 85% training accuracy, 65% testing accuracy and overall average precision of 58%. Dyslexia prediction was done using KNN, yielding a fruitful accuracy of around 89%. Furthermore, testing was done for a group of 65 children, 25 of them were identified as true positive, 4 false negative, 36 true negative while no one was identified as false positive exhibiting the success of screening.

B. Letter dysgraphia screening

Initially, handwritten images gathered from normal children were normalized to gain fixed-size images of 50 * 50 pixels. Screening was done for Sinhala letters and for each letter 5000 images were gathered. Segmentation was done to separate words into individual letters. Training accuracy of 95% and 85% test accuracy was achieved by the neural network used for screening of letter dysgraphia. 90% accuracy was achieved for predicting letter dysgraphia using SVM algorithm. From a group of 60 students, 21 students were marked as true positive, 3 students as false negative, 36 students as true negative while no student was marked as false positive.

C. Dyscalculia screening

Screening accuracy is used as a benchmark to measure the performance of the model. Using an SVM model, an accuracy of 92% was reached in classifying children with dyscalculia. Random forest and naive bayes both gave an accuracy of 88%. Therefore, it can be seen that SVM is the most effective algorithm in classifying dyscalculia.

D. Numeric dysgraphia screening

In classifying data into two classes as dysgraphic or not five algorithms were used to check the best option namely RF, SVM, KNN, logistic regression, decision tree and they gave testing accuracy of 92%, 88%, 84%, 87% and 88% respectively. From the experimental results obtained, it can be concluded that RF is more appropriate for predicting numeric dysgraphia as it showed an accuracy of 92%.

V. CONCLUSION

There are many applications designed for screening and intervention of these learning disabilities, but the vast majority of these applications are in English and do not specifically focus on primary age children between the age of 6 to 7. We have presented a mobile application in Sinhala that provides screening with highly accurate results as well as intervention without the need for separate applications.

In the future, we plan to explore support for multiple languages and mobile operating systems. Also, gathering data from children across multiple age groups to improve the existing datasets is part of the planned future work.

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